# IBM HR ANALYTICS

## EMPLOYEE ATTRITION ANALYSIS

*by Rudijanto H, LHUB Capstone Project*

*Understanding employee turnover patterns to support better HR decision and employee retention*

**Employee Retention**
Understand why retaining employee matters

**Attrition Patterns**
Identify differences across employee group

**HR Insights**
Identify areas for further investigation

*Analytical Approach*

- *Descriptive Analysis*
- *Predictive Analysis*

*Tools: Google Colab, Python, Power BI*

# BUSINESS PROBLEM

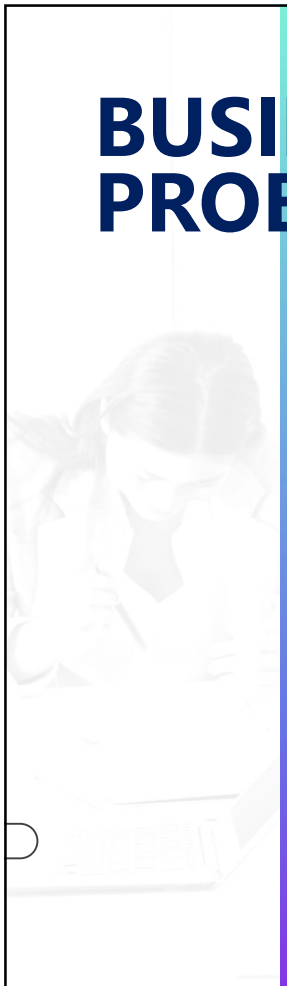
*In today's fast-paced business environment, organisations collect large amounts of data to understand business performance and support decision-making. However, having data alone is not enough. It must be analysed effectively to identify meaningful patterns, understand potential problems and uncover areas that require attention.*

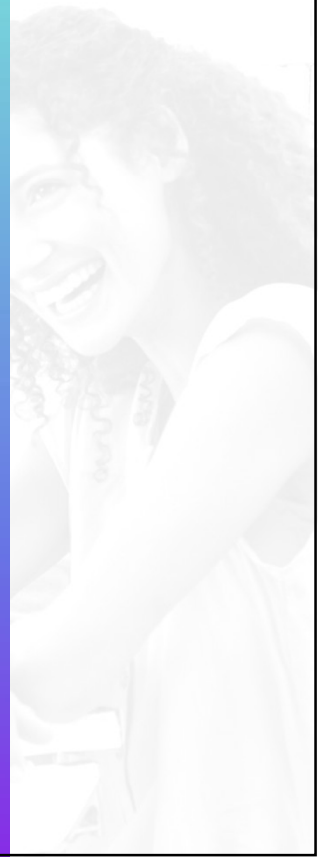
## Project Overview

*Employee attrition can increase recruitment and training costs, disrupt team stability and affect productivity. This project focuses on descriptive analysis of IBM HR Analytics data to understand employee attrition patterns across departments, job roles, compensation, employee satisfaction, overtime and years of service. The findings are presented through an interactive Power BI dashboard to support HR decision-making and workforce planning.*

## Problem Statement

*How can a company identify the key factors linked to employee turnover and use predictive analysis to identify employees who may be more likely to leave, so that it can reduce staff turnover, lower costs, increase productivity and support better workforce planning?*





## COMPENSATION

*To analyse the relationship between employee salary, compensation and attrition to understand how compensation levels are associated with employee turnover.*

## DEPARTMENT AND JOB ROLE

To identify departments and job roles with higher employee attrition rates and examine differences in turnover across employee groups.

## SATISFACTION AND WORK ENVIRONMENT

*To examine the relationships between job satisfaction, work environment, work-life balance and employee attrition.*

## OVERALL ATTRITION

*To measure the overall employee attrition rate and compare employees who leave with those who remain in the organisation.*



## OVERTIME, WORKING EXPERIENCE AND YEARS OF SERVICE

*To investigate the relationships between overtime, work experience, years of service and employee attrition to identify workforce patterns associated with employee turnover.*

# RESEARCH OBJECTIVES

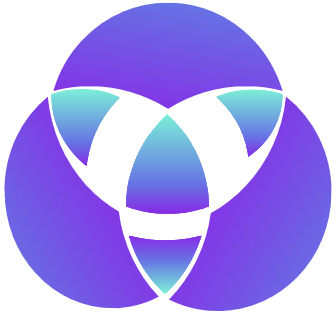
# Data Source & Preparation

### Data Source

- *Employee*
- *Compensation*
- *Employment Status*
- *Experience*
- *Performance*

### Preparation

- *Combined using EmployeeNumber*
- *Checked missing values and data types*
- *Created analysis categories*
- *Prepared data for analysis and modelling*



**Five source CSV files**
Employee · Compensation · Employment Status · Experience · Performance

↓

**Combine the datasets**
Match employee records using EmployeeNumber

↓

**Clean and prepare**
Check missing values, column names and data types

↓

**Create derived categories**
Income Band · Experience Group · Service Group

↓

**Final Prepared Dataset**
**1,470 records**
Used for descriptive analysis, predictive modelling and Power BI

# Methodology

**Data preparation and quality checks**
Load, inspect, clean and combine the five datasets.

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**Descriptive Analysis**
Identify attrition patterns across job roles, income, satisfaction, overtime and service

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**Compare employee groups**
Use tables and charts to examine RO1–RO5.

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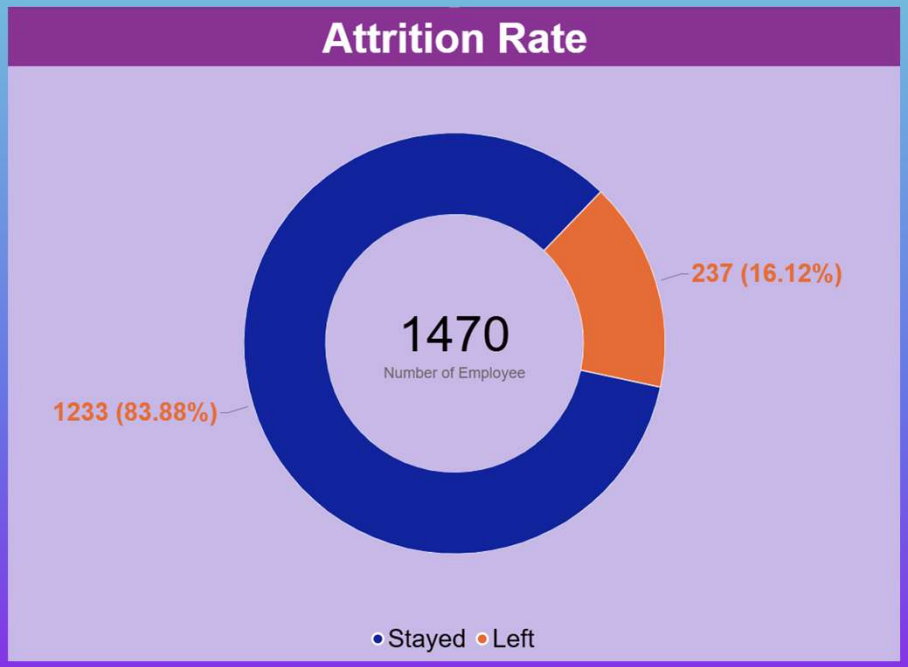
**Power BI Visualisation**
Use Power BI to communicate the results through an interactive dashboard.

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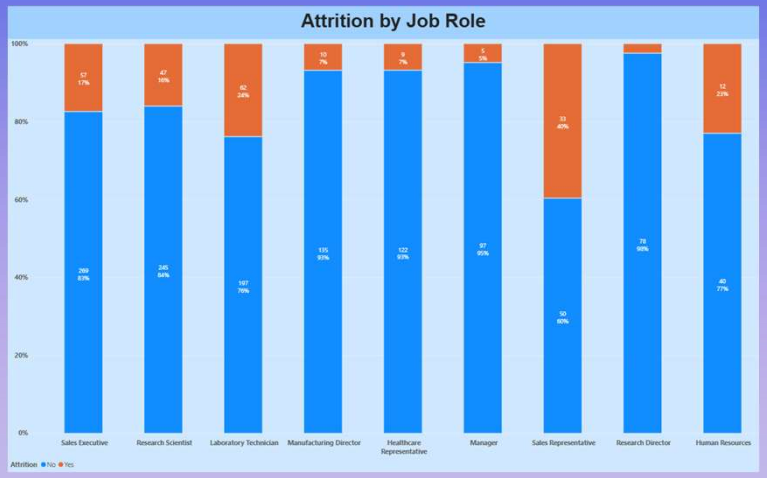
**Conclusion & HR Recommendations**
Connect the evidence to practical HR priorities.

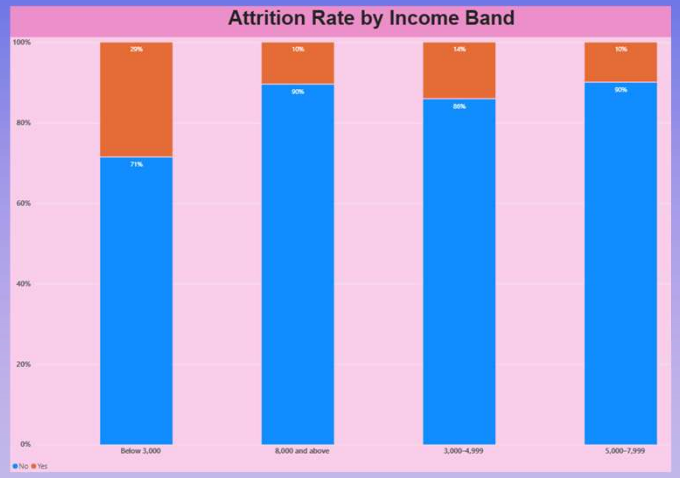
# OVERALL EMPLOYEE ATTRITION

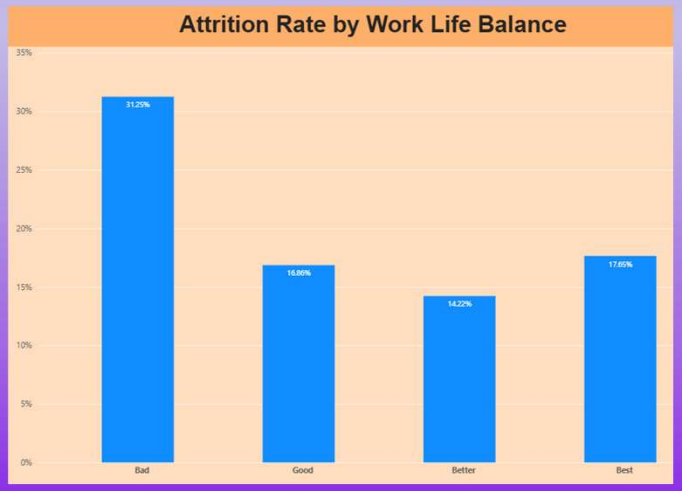
| Total Employees | Attrition Rate |
| --- | --- |
| 1,470 | 16.12% |

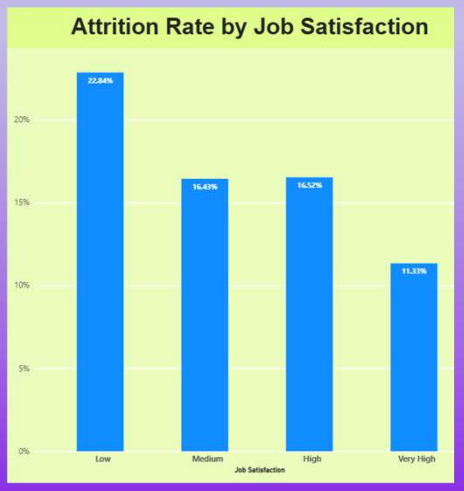


# KEY DESCRIPTIVE FINDINGS



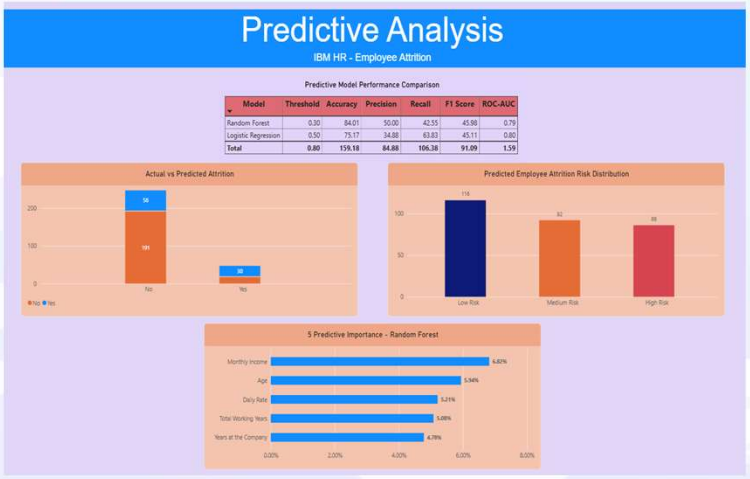



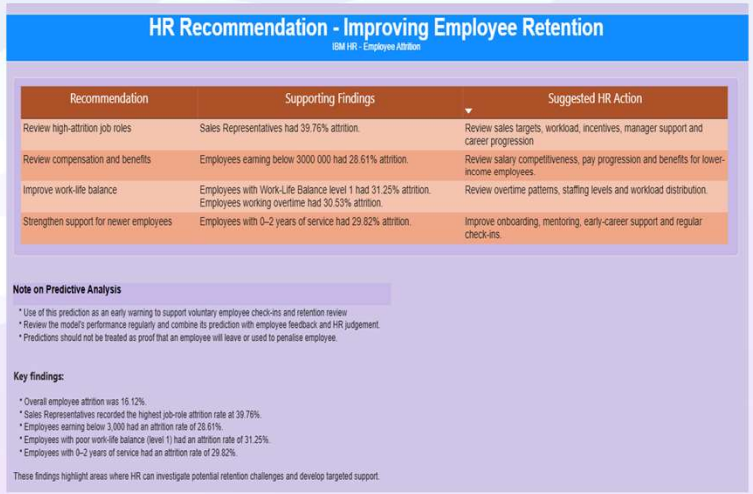



# POWER BI DASHBOARD

from Data to Insights



# HR Recommendation

***1. Review Job Roles and Department***

*Sales Representatives showed the highest job-role attrition rate at 39.76%. HR could review workload, career development, targets and incentives.*

***2. Review Compensation***

*Employees earning below 3,000 had a 28.61% attrition rate. HR could review pay competitiveness, benefits and salary progression.*

***3. Improve Employee Support***

*Lower work-life balance and satisfaction ratings were associated with higher attrition. HR could review workload, employee feedback and support programmes.*

***4. Support Overtime and New Employees***

*Attrition was 30.53% among employees working overtime and 29.82% among those with 0–2 years of service. HR could review staffing, onboarding and mentoring.*

***Predictive analysis provides additional estimates of attrition risk to support further investigation***

# Conclusion:

**Key conclusions**

- Attrition varied across job roles, income groups, satisfaction levels, overtime status and years of service.
- Descriptive analysis helped identify employee groups with higher attrition rates.
- Power BI brought the findings together in an interactive dashboard.
- Predictive modelling provided an additional way to estimate attrition risk.

**Limitations and next steps**

- The dataset reflects historical records and may not represent a current workforce.
- Patterns show associations, not proof of cause and effect.
- Future work could include updated employee data, employee feedback and evaluation of retention initiatives.